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APPARATUS AND METHODS FOR JOINING BONES

Abstract

Apparatus and methods are described for use in securing a first bone relative to a second bone using a pair of surgical staples or, alternatively, a surgical staple and a bone screw. When a pair of staples are used, one of the legs of one of the staples may pass between the legs of the another of the staples. When a staple and a bone screw are used, the bone screw may pass between the legs of the staple. A guide tool can be used during the drilling of the holes for the legs of the staples or the legs of the staple and a bone screw. The guide tool provides guides for drilling guide holes for the staples or for the screw and staple.

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Background/Summary

FIELD

[0001] Apparatus and methods for joining or fusing one bone or two bones together are described herein and, more specifically, apparatus and methods for drilling guide holes in bones for receiving surgical staples or a surgical staple and a bone screw.

BACKGROUND

[0002] Screws are commonly used for joining or fusing two bones or bone pieces together. The screw or screws may cross a joint, fracture or osteotomy. For example, screws can be used to fuse metatarsal phalangeal (MTP) joints to relieve pain. In another example, screws can be used in a Lapidus procedure to fuse the joint between the first metatarsal bone and the medial cuneiform. Instead of screws, fusions can be made using compression staples. A problem with using screws or staples, particularly where multiple such fasteners are used in a single procedure, is that often guide holes for receiving the screws or legs of the staples are drilled free-hand. Although suitable fusions can be made using holes that are drilled free-hand, this can be time consuming and potentially prone to errors in the relative positioning and orientating of the staple legs or screw.

[0003] The guide tools and methods described herein can address these problems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a perspective view of an embodiment of a guide tool for use in drilling holes in bones for insertion of a first surgical staple and a second surgical staple, the guide tool having a set of rearward guides for use in drilling holes to receive legs of the first surgical staple and a moveable set of forward guides for use in drilling holes to receive legs of the second surgical staple;

[0005] FIG. 2 is a top plan view of the guide tool of FIG. 1;

[0006] FIG. 3 is a left side elevation view of the guide tool of FIG. 1;

[0007] FIG. 4 is a right side elevation view of the guide tool of FIG. 1;

[0008] FIG. 5 is a front side elevation view of the guide tool of FIG. 1;

[0009] FIG. 6 is a rear perspective view of a movable bracket of the forward guides of the guide tool of FIG. 1;

[0010] FIG. 7 is a detailed view of the front end of the handle of FIG. 1;

[0011] FIG. 8 is an exploded view of the guide tool of FIG. 1;

[0012] FIG. 9 is a perspective view of an operative portion of the guide tool of FIG. 1 positioned adjacent a bone in after guide holes have been drilled in the bone and staples inserted into the guide holes;

[0013] FIG. 10 is another perspective view of an operative portion of the guide tool of FIG. 1 positioned adjacent a bone in before guide holes have been drilled in the bone and before staples inserted into the guide holes;

[0014] FIG. 11 is perspective view, similar to that of FIG. 10, of an operative portion of the guide tool of FIG. 1 positioned adjacent a bone in after some but not all of the guide holes have been drilled in the bone and before staples are inserted into the guide holes;

[0015] FIG. 12 is a perspective view of a pair of staples having been inserted into holes drilled in the bones;

[0016] FIG. 13 is a perspective view of a second embodiment of a guide tool for use in drilling

holes in bones for insertion of a surgical staple and a bone screw, the guide tool having a set of rearward guides for use in drilling holes to receive legs of the surgical staple and a moveable forward guide for use in drilling a hole to receive the bone screw, the guide tool being positioned adjacent bones for drilling holes;

[0017] FIG. **14** is a different perspective view of the tool of FIG. **13** positioned adjacent bones for drilling holes, showing a guide wire inserted through a pin disposed in the forward guide;

[0018] FIG. **15** is a perspective view of a staple and a screw having been inserted into holes drilled in the bones;

[0019] FIG. **16** is a perspective view of an alternative embodiment of a guide tool for use in drilling holes in bones for insertion of a first surgical staple and a second surgical staple, the guide tool having a set of rearward guides for use in drilling holes to receive legs of the first surgical staple and a moveable set of forward guides for use in drilling holes to receive legs of the second surgical staple, and having multiple openings through which guide pins can be inserted and passed into adjacent bones for positing the tool prior to the drilling of some or all of the guide holes;

[0020] FIG. **17** is a perspective view of the tool of FIG. **16**, showing guide holes having been drilled for the legs of the staples;

[0021] FIG. **18** is another alternative tool, having a portion that can clamp or at least partially anchor onto an already-inserted staple, and a forward portion with a pair of drilling guides;

[0022] FIG. **19** is another alternative tool, having a portion that can clamp or at least partially anchor onto an already-inserted staple, and a forward portion with an offset drilling guide for drilling a guide hole for a screw;

[0023] FIG. **20** is a perspective view of a surgical staple having two legs;

[0024] FIG. **21** is a side elevation view of the surgical staple of FIG. **20**, showing legs of the staple at acute angles relative to a bridge;

[0025] FIG. **22** is a perspective view of a surgical staple having four legs;

[0026] FIG. **23** is a side elevation view of the surgical staple of FIG. **22**, showing legs of the staple at acute angle relative to a bridge;

[0027] FIG. **24** is a perspective view of a surgical staple having four legs;

[0028] FIG. **25** is a side elevation view of the surgical staple of FIG. **22**, showing legs of the staple at acute angle relative to a bridge;

[0029] FIG. **26** is a perspective view of yet an alternative embodiment of a guide tool for use in drilling holes in bones for insertion of a first surgical staple and a second surgical staple, the guide tool having a set of rearward guides for use in drilling holes to receive legs of the first surgical staple and a fixed set of forward guides for use in drilling holes to receive legs of the second surgical staple;

[0030] FIG. **27** is a side elevation view of the guide tool of FIG. **26**;

[0031] FIG. **28** is a perspective view of another alternative embodiment of a guide tool for use in drilling holes in bones for insertion of a surgical staple and a bone screw, the guide tool having a rearward set of guides for use in drilling holes to receive legs of the surgical staple and a set for forward guides for use in drilling a hole to receive the bone screw;

[0032] FIG. **29** is a different perspective view of the guide tool of FIG. **28**;

[0033] FIG. **30** is a top plan view of the guide tool of FIG. **28**;

[0034] FIG. **31** is a top plan view of an alternative to the guide tool of FIG. **28**, the with the forward guides being at a different angle compared to those of the guide tool of FIG. **28**;

[0035] FIG. **32** is a perspective view of an alternative embodiment of a guide tool for use in drilling holes in bones for insertion of a first surgical staple and a second surgical staple, the guide tool having a set of rearward guides for use in drilling holes to receive legs of the first surgical staple and a moveable set of forward guides for use in drilling holes to receive legs of the second surgical staple;

[0036] FIG. **33** is a left side elevation view of the guide tool of FIG. **32**;

[0037] FIG. **34** is a top plan view of the guide tool of FIG. **32**;
[0038] FIG. **35** is a partial section view of the guide tool of FIG. **32** taken along line **35-35** of FIG. **34**, showing a bracket in one position and locking knob of the bracket in a first, clamping position;
[0039] FIG. **36** is a partial section view of the guide tool of FIG. **32**, similar to that of FIG. **35** but showing the bracket in a second position and the locking knob of the bracket in a second, unclamping position;
[0040] FIG. **37** is an exploded perspective view of the moveable set of forward guides of the tool of FIG. **32**;
[0041] FIG. **38** is a perspective view of a segment of a handle for a surgical tool, with the hatching showing that the handle could be used with any surgical tool;
[0042] FIG. **39** is a top plan view of the handle of FIG. **38**;
[0043] FIG. **40** is a right side elevation view of the handle of FIG. **38**, the left side elevation view being the mirror image thereof;
[0044] FIG. **41** is a front elevation view of the handle of FIG. **38**;
[0045] FIG. **42** is a bottom plan view of the handle of FIG. **38**; and
[0046] FIG. **43** is a rear elevation view of the handle of FIG. **38**.

DETAILED DESCRIPTION

[0047] As described herein and shown in FIGS. **1-37**, apparatus and methods are disclosed for use in securing a first bone relative to a second bone using a pair of surgical staples or, alternatively, a surgical staple and a bone screw. When a pair of staples are used, advantageously at least one of the legs of one of the staples passes between the legs of the another of the staples. If the legs of the staple are coplanar, then this can beneficially result in intersecting planes of securement. When a staple and a bone screw are used, advantageously the bone screw passes between the legs of the staple. This can beneficially result in the bone screw intersecting a plane of the legs of the staple. A guide tool can be used during the drilling of the holes for the legs of the staples or the legs of the staple and a bone screw. Advantageously, the guide tool provides guides for drilling the guide holes such that the guides can be fixed relative to each other to improve accuracy during drilling and make the fusion procedure faster, simpler and readily reproducible.

[0048] A first embodiment of the guide tool **10** includes a rearward set of drill guides **12** fixed relative to a handle **14** and an adjustable forward set of drill guides **16** that can be selectively fixed relative to the rearward set of drill guides **12**, aspects of which are shown in FIGS. **1-12**. The guide tool **10** can be used for drilling guide holes in bones for insertion of a legs **102** of a first surgical staple **100** and legs **112** of a second surgical staple **110**. The rearward drill guides **12** can be used to drill a plurality of parallel guide holes for receiving the legs **102** of the first staple **100** which, in the exemplary embodiment, has four legs **102** generally in a first staple plane, such as those shown in FIGS. **22-25**. The forward drill guides **16** can be used to drill a plurality of parallel holes for receiving the legs of the second staple which, in the exemplary embodiment, has two legs generally in a second staple plane, such as shown in FIGS. **20** and **21**. The guide tool **10** is configured such that the holes are aligned so that the first and second staple planes intersect once the staples are inserted into the bones, as shown in FIG. **12**.

[0049] Broadly, the guide tool **10** includes the handle **14** with an attached rearward set of drill guides **12**. An adjustable arm **18** has a linear segment **20** that can be selectively attached to the handle **14** in a variety of different positions. The adjustable arm **18** also has an arcuate segment **22** that is spaced from the handle by the linear segment of the arm. A bracket **24** is slidable along the arcuate segment **22** and carries the forward set of drill guides **16**. The bracket, and thus the forward set of drill guides, can be selectively fixed relative to the arcuate segment of the arm.

[0050] Each of the drill guides **12**, **16** includes a cylindrical sleeve **26** having a through-opening extending along a central axis thereof. In use, a drill bit can be inserted into the through-opening and used to drill a hole. The through-opening and drill bit are preferably sized so that play between the drill bit and the sleeve is minimized so that a hole can be drilled with accuracy. The drill guides

also include a separate holder **28** for each of the cylindrical sleeves. Each of the holders includes a through-bore for receiving part of one of the cylindrical sleeves. Each of the cylindrical sleeves can optionally have a serrated edge at the distal tip thereof for seating on an adjacent bone.

[0051] The cylindrical sleeves are axially and rotationally secured in the through-bores of the holders. Each of the holders has a pair of aligned apertures tending perpendicular relative to a longitudinal axis passing through a center of the holder. An inner circumferential wall of the holder has partially dished in alignment with the aligned apertures. Each of the cylindrical sleeves also has a dished portion on an outer circumference thereof, extending generally perpendicular relative to the central axis thereof. When one of the cylindrical sleeves is received with the through-bore of one of the holders, the dished portion of the cylindrical sleeve is positioned to face the dished portion of the inner circumferential wall of the holder such that an opening of circular cross-section is formed therebetween. When so arranged, a pin can be positioned with each end being held by the aligned apertures and through the opening formed by the dished portions. The pin is thus fixed to the holder via the aligned apertures thereof. Engagement between the fixed pin and the dished portion of the cylindrical sleeve both prevents rotation of the cylindrical sleeve within the holder and axial movement of the cylindrical sleeve within the holder.

[0052] The handle **14** includes a longitudinally extending grasping portion, which can optionally be ergonomically contoured for being held by the hand of a surgeon or other user. As mentioned above, the rearward set of drill guides **12** are attached to the handle **14**. More specifically, there are four drill guide holders integrally formed at one end of the handle. The four holders are arranged in a line that is generally perpendicular to the grasping portion of the handle, as shown in FIG. 2. Other arrangements of the holders relative to the handle can be made, e.g., parallel or angled; and the holders do not have to be in a line, but can be in other non-linear arrangements. In the illustrated embodiment, there are four holders. It will be understood that a different number of holders can be used depending upon the number of staple legs to be inserted into holes, e.g., two, three, etc.

[0053] The handle includes an open channel **30** formed between the four holders, as seen in FIG. 7. The channel is configured to receive the linear segment of the adjustable arm. More specifically, two of the holders are on one side of the channel and two of the holders on another side of the channel. The channel has a flat bottom wall, a rear end with an obround opening **32**, and a projecting end **34** extending forward of the holders. The linear segment of the adjustable arm terminates in a protuberance that is obround in cross-section and keyed to fit within the oval-shaped opening of the rear end of the channel of the handle. A flattened bottom of the protuberance abuts the flat bottom wall of the channel. Accordingly, engagement between the channel and the protuberance of the linear segment of the arm restricts rotation of the arm relative to the handle.

[0054] The arm can be selectively attached relative to the handle, as mentioned above. To that end, a threaded screw or knob **40** can be used to secure the arm to the handle. More specifically, the linear segment of the adjustable arm has an elongated slot **42** formed therein. When the protuberance of the linear segment of the arm is received within the channel of the handle, the slot is positioned above a threaded bore **44** formed in the projecting end of the channel. The knob has a threaded shaft depending from a head with a knurled periphery. The threaded shaft of the knob is inserted through the slot of the linear segment of the arm and into the threaded bore. The tightening of the knob can clamp the arm to the handle. The knob can be removed to either detach the arm from the handle or to loosen to allow the arm to be slidably adjusted relative to the handle and then tightened.

[0055] The bracket **24**, which carries the forward set of drill guides, is slidable along the arcuate segment. A pair of the holders are provided on the bracket, one on each side of the arcuate bottom wall thereof. The arcuate segment has an arcuate, radially outward surface and an arcuate, radially inward surface. The arcuate segment has a generally rectangular cross-section. The bracket includes an arcuate bottom wall **25**, as shown in FIG. 6, with a pair of lateral, upstanding side walls

27. A radius of curvature of the bottom wall of the bracket generally matches a radius of curvature of the arcuate segment of the arm such that the bottom wall of the bracket can slide along the outward surface of the arcuate segment of the arm. Each of the pair of sidewalls has three apertures. The apertures of one of the pair of sidewalls is aligned with the apertures of the other of the pair of sidewalls. Pins **41** are inserted through aligned apertures, such that there are three pins. The apertures, and thus the pins, are aligned such the dowels can slide along the inward surface of the arcuate segment of the arm. The pins and the arcuate bottom wall sandwich the radially inward surface and the radially outward surface of the arcuate segment, respectively, therebetween.

[0056] The pair of sidewalls and the bottom walls optionally have windows formed therein through which indicia on the arcuate segment of the arc can be visible to help with positioning the bracket along the arcuate segment of the arm.

[0057] The bracket, and thus the forward set of drill guides, can be selectively fixed relative to the arcuate segment of the arm, as mentioned above. The arcuate segment of the adjustable arm has a generally rectangular cross-section. One side of the arcuate segment has a curved, obround-shaped groove **43**. One of the sidewalls of the bracket has an extending ear with a threaded through-bore. A knob **45** is provided to secure the bracket on the arcuate segment of the arm. More specifically, the knob has a threaded shaft depending from a head with a knurled periphery. The threaded shaft of the knob is threaded into the threaded through-bore of the ear of the sidewall of the arcuate segment of the arm and into engagement with a bottom of the groove on the arcuate segment of the arm. The tightening of the knob can clamp the bracket to the arm. The knob can be removed to allow the bracket to be slid off the arm or loosened to allow the bracket to be slidingly adjusted relative to the arm and then tightened to fix the bracket in position relative to the rearward set of drill guides. Ends of the groove on the arcuate segment of the arm can limit the extent to which the bracket can slide on the arm when then threaded shaft is received within the groove but not engaged with the bottom wall thereof.

[0058] Turning now to a method of using the first embodiment of the guide tool, the handle can be used to position the rearward set of drill guides against a pair of adjacent bones, as shown in FIG. **10**. The adjacent bones can be separate bones or pieces of the same bone and, as used herein, the term bones as used herein is understood to include both two different bones or pieces of the same bone or the same bone even if not into separate pieces. In other words, even if used in the plural sense the term can encompass the singular sense.

[0059] The forward set of drill guides can also be positioned against the pair of adjacent bones, also as shown in FIG. **10**, with adjustment to the position of the forward set of drill guides being made using the arm and/or the bracket. As used herein, adjacent bones can also include bone pieces of the same bone that require fusion. Optionally, a guide wire for a cannulated screw can be installed in one of the bones. When used, the guide wire can then be threaded through one of the rearward set of drill guides to help with positioning the guide tool adjacent the bones. A first of the holes can then be drilled using a drill bit—attached to a drill—inserted through the first of the holes. A surgeon can drill the remainder of the holes or, optionally, insert one or more removable pins temporarily through the drill guides to temporarily secure the guide tool to the bone or bones. For example, a hole can be drilled into one of the pair of bones using one of the rearward set of drill guides to guide the drill bit; and another hole can be drilled into the same or a different one of the pair of bones using the forward set of drill guides to guide the drill bit, as shown in FIG. **11**. Removable pins can be inserted into those holes to temporarily secure the guide tool to the bone or bones. The remainder of the holes can be drilled, after which the pins can be removed and the guide tool removed from adjacent the bones. Staples can then be inserted into the holes that are drilled into the bones to fuse the bones together, as shown in FIG. **12**.

[0060] Preferably, though not necessarily, the legs of each of the staples in their resting or unbiased state are at acute angles relative to a bridge that joins the legs, as shown in FIGS. **20-25**. The staples can optionally be made of a shape memory metal, such as nitinol. The drilled holes made

using the rearward set of drill guides are perpendicular to each other, as are the drilled holes made using the forward set of drill guides. The legs of the staples can be temporarily bent to be generally perpendicular relative to the bridges and generally parallel relative to each other and, when in that arrangement, inserted into the holes. An insertion tool, such as that disclosed in U.S. patent application Ser. No. 17/322,580, filed May 17, 2021, which is hereby incorporated herein by reference in its entirety, can engage with the staple to bend and temporarily hold the legs in the perpendicular and parallel arrangement for insertion into the holes. Once the legs are almost completely inserted, the insertion tool can be disengaged from the staple and the staple legs inserted the rest of the way into the holes. Advantageously, the shape memory properties of the staple can cause the legs to want to return to their acutely angled orientation relative to the bridge, thereby compressing the adjacent bones together, preferably with a compressing force that is greater than if the legs in their resting or neutral state were generally perpendicular relative to the bridge.

[0061] In a guide tool **210** of a variation of the first embodiment of the guide tool, shown in FIGS. **32-37**, a bracket **224**, which carries the forward set of drill guides **216**, is slidable along an arcuate segment **219** and can be selectively fixed in one of a plurality of different indexed positions. The method of using the variation of the first embodiment can be the same as for the first embodiment, described herein, with the difference being that the forward set of drill guides **216** can be selectively fixed in one of a plurality of different preset, indexed positions. The bracket **224** can include apertures for use with insertion of a guide wire for securing the bracket **224** and, if clamped, the arm **218**, relative to the bone or bones.

[0062] The bracket **224** of the variation of the first embodiment is otherwise generally similar in construction to the bracket **24** of the first embodiment, with a pair of sidewalls **227** are provided on the bracket **224**, one on each side of the arcuate bottom wall thereof. The arcuate segment has an arcuate, radially outward surface and an arcuate, radially inward surface. The arcuate segment has a generally rectangular cross-section. The bracket includes an arcuate bottom wall **225**, as shown in FIGS. **35-37**, with a pair of lateral, upstanding side walls **227**. A radius of curvature of the bottom wall **225** of the bracket **224** generally matches a radius of curvature of the arcuate segment **219** of the arm **218** such that the bottom wall **225** of the bracket **224** can slide along the outward surface of the arcuate segment **219** of the arm **218**. Each of the pair of side walls **227** has two apertures. The apertures of one of the pair of side walls **227** is aligned with the apertures of the other of the pair of sidewalls **227**. Pins **241** are inserted through aligned apertures, such that there are two pins **241**. The apertures, and thus the pins **241**, are aligned such the pins **241** can slide along the inward surface, i.e., the underside, of the arcuate segment **219** of the arm **218**. The pins **241** and the arcuate bottom wall **225** sandwich the radially inward surface **237** and the radially outward surface **239** of the arcuate segment **219** of the arm **218**, respectively, therebetween.

[0063] The bracket **224**, and thus the forward set of drill guides **216**, can be selectively fixed relative to the arcuate segment of the arm, as mentioned above, in one of a plurality of preset indexed positions. The arcuate segment **219** of the adjustable arm **218** has a plurality of detents **243** on the radially outward surface **239** thereof, as shown in FIGS. **35-37**. The bracket **224** carries a locking pin **252** that can be partially received in one of the plurality of detents **243** and locked or fixed so that the bracket **224**, and thus the forward set of drill guides **216**, is fixed in one of a plurality of present positions along the arm **218**.

[0064] More specifically, the arcuate bottom wall **225** of the bracket **224** has an upstanding stem **260**, shown in FIGS. **35-37**, with a central opening extending through the stem **260** and the bottom wall **225** of the bracket **224**. The locking pin **252** is slidably received with the central opening of the stem **260**. One end of the locking pin **252** is positioned to be received in one of the plurality of detents **243**, as shown in FIG. **35**. The other end of the locking pin **252** extends through the central opening of the stem **260**.

[0065] A knurled locking knob **270** is attached to the stem **260** and has two positions, as will be

explained further herein. The knurled locking knob **270** has an inner bore for receiving a second end portion of the locking pin **252**. The inner bore also has an inner thread that can threadingly engage an outer thread **262** of the upstanding stem of the bracket **224**. A spring **266** surrounds the second end portion of the locking pin **252** and biases the locking pin **252** into engagement with one of the plurality of detents **243** of the arm **218**. The locking knob **270** can be rotated clockwise or counterclockwise to axially move the knob **270** along the stem **260** either away from the bottom wall **225** of the bracket **224**, with counterclockwise rotation, or toward the bottom wall **225** of the bracket **224**, with clockwise rotation.

[0066] In a first, clamping position, shown in FIG. **35**, the locking knob **270** is rotated clockwise until the locking pin **252** is clamped between the locking knob **270** and the arm **218**, thereby fixing the bracket **224**—and thus the forward set of drill guides **216**—relative to the arm **216**. In a second position, shown in FIG. **36**, the locking knob **270** is rotated counterclockwise until the locking pin **252** is no longer clamped and the bracket **224** can slide along the arm **218**. When in the second position, the spring **266** biases the locking pin **252** away from the arm **218**. However, as the bracket **224** slides along the arm **218**, the locking pin **252** can axially move within the inner bore of the locking knob **270**. When sliding along the arm **218**, when the one end of the pin **252** engages with one of the detents **243** a small force from the spring **266** is required to move the one end of the pin **252** out of engagement with the one of the detents **243**. A user can feel that a force is required to slide the bracket **224** further once the pin **252** has engaged with the one of the detents **243**, thereby providing tactile feedback that one of a plurality of indexed positions of the bracket **224** along the arm **218** has been reached. The plurality of preset positions can be selected according to the particular needs of the guide tool **210**, and can, for example, be at various angles along the arm **218** of between 30 and 150 degrees, for example, 30, 45, 60, 75, 90, 105, 120, 135 and 150 degrees, or other increments therebetween, e.g. increments of 5, 7.5, 10, etc. degrees.

[0067] The axial position of the locking knob **270** relative to the stem **260** of the bracket **224** when in the second position is limited. More specifically, the stem **260** includes an annular groove **264** positioned between the outer thread **262** and the bottom wall **225** of the bracket **224**. The locking knob **270** includes a locking bore **272** passing through the central bore but offset relative to a central axes of the central bore. A fixation pin **274** can be received with the locking bore **272** of the locking knob **270**. The fixation pin **274** is positioned such that the it can axially move up and down within the annular groove **264** when the locking knob **270** is rotated to move between the first and second position, and vice versa, However, when the fixation pin **274** reaches an upper limit of the groove **264**, at the intersection with the outer threads **262**, further counterclockwise rotation of the locking knob **270** is prevented by engagement of the fixation pin **274** with the intersection with the outer threads **262** where there is a larger diameter compared to that of the annular groove **264**.

[0068] The arm **218** includes an upper stop, in the form of a step in the arm **218**, and a lower stop, in the form of a pin received in an aperture of the arm **218**, as shown in FIGS. **32**, **33**, **35** and **36**. The upper and lower stops limit the extend to which the bracket **224** can slide along the arm **218**.

[0069] Another variation of the first embodiment of the guide tool is depicted in FIGS. **26** and **27**, with like numbers referencing like parts. The guide tool **111** depicted in FIGS. **26** and **27** includes a rearward set of drill guides **112** fixed relative to a handle **114** and a forward set of drill guides **116**. This variation differs from those of FIGS. **1-12** and **32-36** in that the forward set of drill guides **116** is fixed. The fixation is permanent, i.e., the forward set of drill guides **116** cannot be adjusted relative to the rearward set of drill guides **112**. Optionally, the rearward set of drill guides **112** and the forward set of drill guides **116** are formed of a unitary body. The handle **114** can also be unitary with the guides **112**, **116**, or separately attached.

[0070] As with the first embodiment and the variation described above, the guide tool **111** of FIGS. **26** and **27** can be used for drilling guide holes in bones for insertion of a legs **102** of a first surgical staple **100** and legs **112** of a second surgical staple **110**. The rearward drill guides **112** can be used to drill a plurality of parallel guide holes for receiving the legs **102** of the first staple **100** which, in

the exemplary embodiment, has four legs **102** generally in a first staple plane, such as those shown in FIGS. **22-25**. The forward drill guides **116** can be used to drill a plurality of parallel holes for receiving the legs of the second staple which, in the exemplary embodiment, has two legs generally in a second staple plane, such as shown in FIGS. **20** and **21**. The guide tool **111** is configured such that the holes are aligned so that the first and second staple planes intersect once the staples are inserted into the bones, as shown in FIG. **12**.

[0071] The guide tool **111** includes the handle **114** with the permanently attached rearward set of drill guides **112**. A forward-extending arm **118** has a first segment **120** and a second segment **122** with a bend **124** therebetween. The forward set of drill guides **116** and the rearward set of drill guides **112** can be positioned such holes made with one set are at a fixed angle relative to the holes made with the other set. This can translate into the first and second staple plates intersecting at a generally corresponding fixed angle. The fixed angle can be, for example, between 30 and 150 degrees, for example, 30, 45, 60, 75, 90, 105, 120, 135 and 150 degrees, or other increments therebetween, e.g. increments of 5, 7.5, 10, etc. degrees. Although an angle of approximately 60 degrees is shown, it will be understood that the tool **111** can be constructed to result in a different angle. For example, a projection **113** along one of the central axes of the guides of the rearward set of drill guides **112** can be considered to intersect a projection **115** along one of the central axes of the forward set of drill guides **116** when viewed laterally, i.e., is a side elevation view as shown in FIG. **27**, although the projections do not actually intersect when viewed from above, e.g., FIG. **26**. The intersection between the two projections **113**, **115** can have an inside angle Θ , as shown in FIG. **27**. The angle Θ can be of any suitable angle, such as, for example, between 30 and 150 degrees, for example, 30, 45, 60, 75, 90, 105, 120, 135 and 150 degrees, or other increments therebetween, e.g. increments of 5, 7.5, 10, etc. degrees, while an angle of about 60 degrees is shown in FIG. **27**.

[0072] Each of the drill guides **112**, **116** includes a cylindrical sleeve **126** having a through-opening extending along a central axis thereof. As described in connection with the prior embodiments, in use, a drill bit can be inserted into the through-opening and used to drill a hole. The through-opening and drill bit are preferably sized so that play between the drill bit and the sleeve is minimized so that a hole can be drilled with accuracy. The drill guides also include a separate holder **128** for each of the cylindrical sleeves. Each of the holders **128** includes a through-bore for receiving part of one of the cylindrical sleeves **126**. Each of the cylindrical sleeves **126** can optionally have a serrated edge at the distal tip thereof for seating on an adjacent bone.

[0073] The cylindrical sleeves **126** are optionally axially and rotationally secured in the through-bores of the holders **128**, as described above.

[0074] The handle **114** includes a longitudinally extending grasping portion, which can optionally be ergonomically contoured for being held by the hand of a surgeon or other user. As mentioned above, the rearward set of drill guides **112** are fixed relative to the handle **114**. More specifically, there are four drill guide holders integrally formed at an end of the handle. There are also four drill guides of the forward set of guides **116**. The holders of each of the respective sets are arranged such that central axes thereof each lie in a common plane, which planes are generally perpendicular to the grasping portion of the handle. Other arrangements of the holders can be made, e.g., parallel or angled; and the holders do not have to be in a line, but can be in other non-linear arrangements. In the illustrated embodiment, there are four holders **128** of the rearward array **112** and two holders **128** of the forward array **116**. A different number of holders can be used depending upon the number of staple legs to be inserted into holes, e.g., two, three, etc.

[0075] The methods of using the variation of FIGS. **26** and **27** can be the same as those of FIGS. **1-12** and **31-36**, with the difference being that the forward set of drill guides **116** are fixed relative to the rearward set of drill guides **112**. Moreover, as shown, the forward set of drill guides **116** and the rearward set of drill guides **112**, and, specifically, the holders thereof, are formed of a unitary body. The unitary body can also include the handle **114**. The unitary body can be cast, injection molded,

machined or otherwise manufactured using a suitable techniques, including being permanently joined together, e.g., welding, fastened, or the like.

[0076] A second embodiment of the guide tool is shown in use in FIGS. **13** and **14**, with like parts having like numbers as compared with the first embodiment of the guide tool. The second embodiment of the guide tool differs from that of the first embodiment in that instead of being used to drill holes for a first staple and a second staple, the guide tool is configured to drill holes for only the legs of one staple and another hole for a bone screw. In particular, the forward set of drill guides is replaced by a single drill guide. Because a bone screw is used instead of one of the staples, the bone screw must be inserted such that it spans between both bones; thus, the hole drilled for the bone screw must also span between both bones. This is accomplished by the length of the bone screw as well as the angle in which the bone screw is inserted. The forward drill guide is angled such that the drill bit drills a hole that is angled, i.e., not perpendicular, to a plane of the holes for the legs of the staple. The linear and arcuate segments of the adjustable arm are at an obtuse angle relative to each other, as shown in FIGS. **13** and **14**, such that a hole drilled using the forward drill guide will be angled to a plane of the holes for the legs of the staple. Like the first embodiment, a guide wire can optionally be used for aligning the guide tool of the second embodiment with the bone or bones. The holes can be sequentially drilled using the drill guides and, optionally, one or more pins can be used for temporarily securing the guide tool to the bone or bones. After drilling of the guide holes, the staple and the screw can be inserted, as shown in FIG. **15**.

[0077] A first variation of the second embodiment of the guide tool is shown in FIGS. **28-30**. Like the second embodiment of the guide tool, the guide tool **300** of the first variation of the second embodiment of the guide tool is configured to drill holes for only the legs of one staple and another hole for a bone screw. The first variation of the second embodiment of the guide tool differs from that of the second embodiment in that it has a forward drill guide **316** that is fixed to an arm **318** such that its angle cannot be adjusted. The arm **318** has four segments, a first segment **320** extending forward from a rear set of drill guides **312** and a handle **314** to a first bend **322**, a second segment **324** extending at an angle relative to the first segment **320** and between the first bend **322** and a second, downward bend **326**, a third segment **328** extending downward from the second bend **326** to a third, inward bend **330**, and a fourth segment **332** extending inward from the third bend **330** to the forward drill guide **316**. Of course, the number of bends and segments can be varied to achieve the same positioning between the drill guides **312** and **316**. For example, the arm can be continuously or partially arcuate. The forward drill guide **316** can include an array of drill guides **317**, such as two or, as shown, three arranged generally vertically relative to each other. Each of the drill guides **317** of the array of drill guides can optionally receive a cannulated guide **319**, as shown in FIGS. **28-30**, for use in insertion of a guide wire. The guide wire can be rotated, in a drilling manner, or otherwise inserted into the bone. Having an array of drill guides **317** allows for a user to select the relative position of the drilled or otherwise formed hole, for example, relative to the drilled holes for the staple made using the rearward drill guides **312**.

[0078] The drill guides **317** optionally each include a lateral opening such that they have generally C-shaped cross section. The lateral opening can be useful during a procedure where a guide wire is used with the forward drill guide **316**. For example, a procedure can include positioning of the tool adjacent one or more bones, then inserting a guide wire into the bone via the drill guides **317** and guide **319**. The guide **319** can then be slid off of the guide wire, opposite the insertion end of the wire, the arm **318** can then be decoupled relative to the handle **314** such that the handle **314** and rearward array of drill guides **312** remain in place. The guide wire can then pass through the lateral opening of the drill guide **317** so that the arm **318** can be removed, leaving the guide wire in place as well as the handle **314** and rearward array of drill guides **312**. The rearward set of drill guides **312** can include apertures for guide wires for use in temporarily fixing the drill guides **312** relative to a bone or bones.

[0079] FIG. **31** shows a second variation of the second embodiment of the guide tool, with like

parts having like numbers. The guide tool **301** difference as compared to the guide tool **300** of the first variation of the second embodiment, shown in FIGS. **28-30**, in that the angle Q between the first and second segments of the arm extends oppositely.

[0080] The methods of using the first and second variation of the second embodiment of the guide tool are the same as described herein with respect to the second embodiment, with the exception that the forward drill guide is fixed to the arm such that the angle cannot be adjusted.

[0081] FIGS. **16** and **17** depict an alternative embodiment of a guide tool for use in drilling holes in bones for insertion of a first surgical staple and a second surgical staple, the guide tool having a set of rearward guides for use in drilling holes to receive legs of the first surgical staple and a moveable set of forward guides for use in drilling holes to receive legs of the second surgical staple. This embodiment differs from the first embodiment in that the handle is of a different size, the arm lacks the extension so that it can be inserted through a rear hole in the channel so that the handle is reversible, and there are multiple openings adjacent the drill guides through which guide pins can be inserted and passed into adjacent bones for positing the tool prior to the drilling of some or all of the guide holes. Any of these features can be incorporated into the other embodiments described herein. FIG. **17** shows guide holes having been drilled for the legs of the staples.

[0082] FIGS. **18** and **19** depict additional alternative tools. Each of these two tools has a portion that can clamp or at least partially anchor onto an already-inserted staple, such as groove or slot for receiving the bridge of the staple. Each of these two tools also includes a forward portion with a pair of drilling guides (FIG. **18**) for drilling guide holes for a staple, in the illustrated embodiment, a two-legged staple, or an offset drilling guide (FIG. **19**) for drilling a guide hole for a screw.

[0083] FIGS. **20-25** disclose various embodiments of staples **100**, **110**, **120**, some staples **110** with two legs **112** (FIGS. **20** and **21**), some staples **100**, **120** with four legs **102**, **122** (FIGS. **22** and **23**), and one **120** with four legs **122**, two of which are shorter than the other two legs (FIGS. **24** and **25**). The legs of the staples are connected via a bridge **104**, **106**, **108**, and the legs can optionally have teeth such as those shown. The use of a staple with two legs shorter than another two legs can beneficially be used for various procedures, such as when one of the bones or the operative portion of the bone has a thickness that may not warrant or be suitable for a longer leg. Other shapes and sizes of staples can be used, such as those disclosed in U.S. patent application Ser. Nos. 29/804,413 and 29/804,411, each filed May 17, 2021, which are each hereby incorporated herein by reference in their entireties.

[0084] The staples are preferably, though not necessarily, made of nitinol or another shape memory material. Sleeves and arm can be made of surgical grade stainless steel, among other materials. The handle and bracket can also be made of surgical grade stainless steel, suitable thermoplastics, e.g., polyphenylsulfone (PPSU), or other suitable materials.

[0085] As shown in the embodiments of FIGS. **28-37**, the arms **218**, **318** and the guides thereon can be removably coupled relative to the handles **218**, **318** and the rearward drill guides associated therewith. This allows for the option during a procedure to decouple the rearward drill guides relative to the forward guide or guides. For example, the tool can be positioned and, optionally secured such as using one or more guide wires, and then the forward guide or guides can be used. The arm carrying the forward guide or guides can then be decoupled from the handle with the handle and associated rearward drill guide remaining in place for use during the procedure. In the illustrated examples, the arm has a rearward portion that engages with a forward portion of the handle. The rearward portion of the arm can be removed downward relative to the forward portion of the handle, or from an opposite side from the direction that the handle projects. Optionally, a protuberance of the rearward portion of the arm extends into an opening fixed relative to the handle. A screw can be used to clamp the arm and handle relative to each other, and the screw can optionally have a knurled knob and be spring biased away from engagement. A shaft of the screw can have an enlarged diameter area for forming a stop against an opening through which the shaft

partially extends. Pins can be used to restrict removal of the screw such that the screw remains attached relative to the handle when the arm is decoupled. A bore in the rearward portion of the arm can include threading for receiving a threaded end of the screw.

[0086] In any of the foregoing embodiments, the components of the tools can optionally include one or more apertures for use with guide wires or the like to attach or position the tool or specific components thereof, such as the handle, rearward set of drill guides and/or forward drill guide or set of guides.

[0087] A handle that can be part of any of the foregoing embodiments is depicted in FIGS. **38-43**.

[0088] All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or language describing an example (e.g., “such as”) provided herein, is intended to illuminate the invention and does not pose a limitation on the scope of the invention. Any statement herein as to the nature or benefits of the invention or of the preferred embodiments is not intended to be limiting. This invention includes all modifications and equivalents of the subject matter recited herein as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context. The description herein of any reference or patent, even if identified as “prior,” is not intended to constitute a concession that such reference or patent is available as prior art against the present invention. No unclaimed language should be deemed to limit the invention in scope. Any statements or suggestions herein that certain features constitute a component of the claimed invention are not intended to be limiting unless reflected in the appended claims. Neither the marking of the patent number on any product nor the identification of the patent number in connection with any service should be deemed a representation that all embodiments described herein are incorporated into such product or service.

Claims

1-20. (canceled)

21. A guide tool for use in drilling holes in bone pieces or bones for insertion of a legs of a first surgical staple and legs of a second surgical staple, the tool comprising: a handle; a rearward set of drill guides fixed relative to the handle; an arm having an arcuate segment; and a forward set of drill guides selectively fixable relative to the arcuate segment of the arm in one of a plurality of preset positions.

22. The guide tool of claim 21, wherein: each of the forward set of drill guides has a longitudinally extending opening with a central axis, through which, in use, a drill bit can be inserted for drilling a hole in an adjacent bone; and each of the rearward set of drill guides has a longitudinally extending opening with a central axis, through which, in use, a drill bit can be inserted for drilling a hole in an adjacent bone.

23. The guide tool of claim 22, wherein one of the forward set and rearward set of drill guides includes four drill guides and another of the forward set and rearward set of drill guides includes two drill guides, and wherein the arm is configured such that the central axes of the two drill guides project between a projection of the central axes of the four drill guides.

24. The guide tool of claim 22, wherein the rearward set of drill guides includes four drill guides and the forward set of drill guides includes two drill guides, and wherein the arm is configured such that the central axes of the drill guides of the forward set of drill guides project between a projection of the central axes of the drill guides of the rearward set of drill guides.

25. The guide tool of claim 21, wherein the forward set of drill guides are carried by a bracket that is slidable along the arcuate segment of the arm.

26. The guide tool of claim 25, further comprising means for selectively fixing the bracket relative to the arcuate segment of the adjustable arm in one of the plurality of preset positions.

27. The guide tool of claim 25, wherein each of the drill guides includes a sleeve with a through-bore for receiving, in use, a drill bit, the sleeves of the rearward set of drill guides being axially and rotationally fixed relative to the handle and the sleeves of the forward set of drill guides being axially and rotationally fixed relative to the bracket.

28. A method of drilling holes in bone pieces or bones for insertion of a legs of a first surgical staple and legs of a second surgical staple using the guide tool of claim 21, the method comprising: positioning the rearward set of drill guides adjacent the bones; positioning the forward set of drill guides adjacent the bones; fixing the position of the forward set of drill guides in one of the plurality of preset positions; inserting a drill bit progressively through each of the drill guides of the rearward set of drill guides and drilling holes in the bones for receiving the legs of the first surgical staple; and inserting a drill bit progressively through each of the drill guides of the forward set of drill guides and drilling holes in the bones for receiving the legs of the second surgical staple.

29. The method of claim 28, wherein at least one of the holes in the bones for receiving the legs of the first surgical staple passes between an adjacent pair of the holes in the bones for receiving the legs of the second surgical staple; or wherein at least one of the holes in the bones for receiving the legs of the second surgical staple passes between an adjacent pair of the holes in the bones for receiving the legs of the first surgical staple.

30. The method of claim 29, further comprising inserting the legs of the first surgical staple into the holes drilled in the bones for receiving the legs of the first surgical staple and inserting the legs of the second surgical staple into the holes drilled in the bones for receiving the legs of the second surgical staple.
